

Six lectures, from ML vocabulary, data structures/preparation to a worked photometric-redshift pipeline.

1a

LECTURE

The Taxonomy of Machine Learning

AI vs ML · paradigms · problem archetypes · why now

1b

LECTURE

Feature Vectors & Data Modalities

frame the problem · X/y · data scales · EDA · curse of dim

2

LECTURE

Data Preparation

missing values · outliers · scaling · encoding · PCA

3a

LECTURE

Splits & Model Reliability

train / val / test · K-fold CV · data leakage

3b

LECTURE

Decision Trees → Random Forests

Gini · pruning · bagging · feature importance · OOB

4

LECTURE

Practical RF + Photo-z Case Study

sklearn pipeline · pitfalls · 10 000 SDSS galaxies

Theme: Foundations → Data preparation → Reliability & models → Practical case study



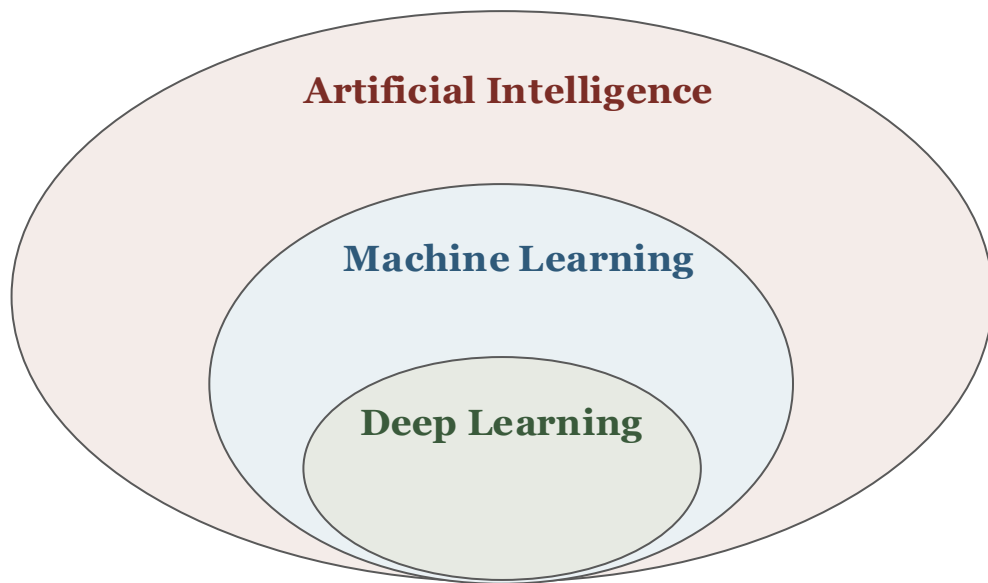
Lecture 1a:

The Taxonomy of Machine Learning

A guided tour of the landscape

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AI – ML – Deep Learning



Artificial Intelligence

The broad pursuit of building systems that perform tasks requiring human-level intelligence.

Machine Learning

A subset of AI focused on statistical algorithms that learn from data — improving with experience.

Deep Learning

ML using deep neural networks — many layers learn hierarchical features automatically.

Today's lecture and the rest of this course are about the middle ring: *classical ML*.

Classification



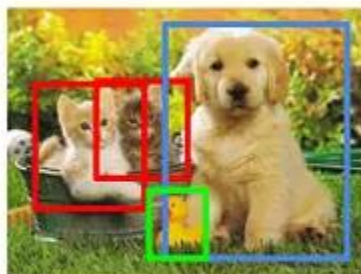
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

**Instance
Segmentation**

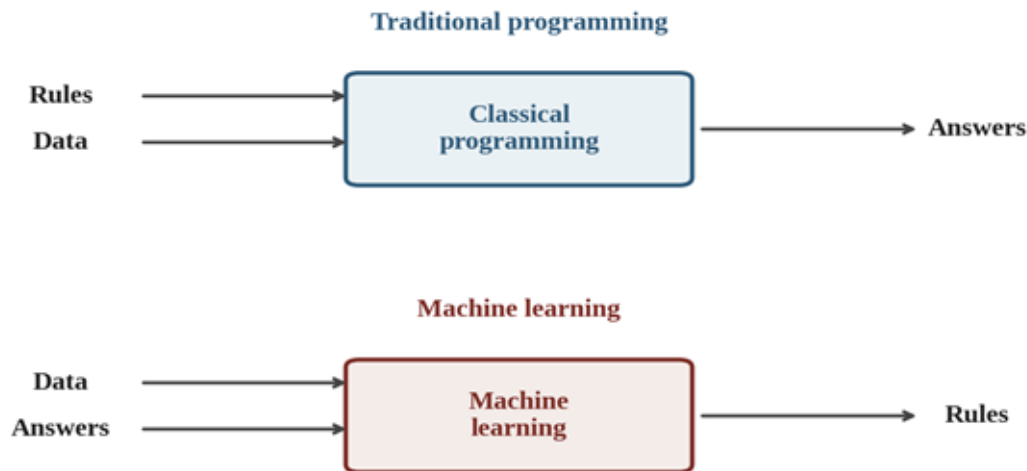


CAT, DOG, DUCK

Single object

Multiple objects

ML vs Traditional Programming



The fundamental flip: in traditional programming we *supply the rules* and let the computer apply them to data. In ML, we *supply data and the answers* and ask the computer to *discover* the rules.

The Four Main Pillars

Machine learning is usually classified by what kind of data the model gets to see.

Supervised

labeled data (X, y)

Learn a mapping $X \rightarrow y$. Classification, regression.

Unsupervised

unlabeled data $(X \text{ only})$

Discover latent structure: clusters, principal components, anomalies.

Semi- / Self-Supervised

a little labeled + lots of unlabeled

Use the structure in unlabeled data to amplify a small labeled set.

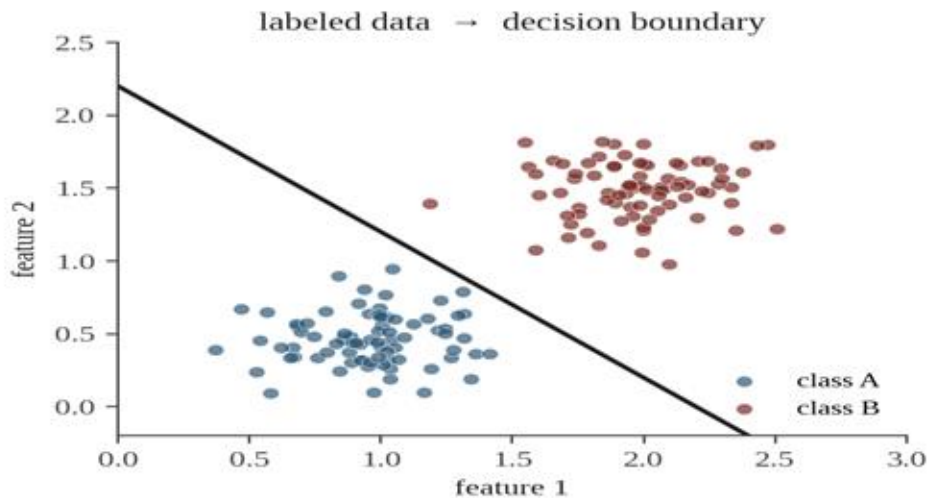
Reinforcement

agent + reward signal

Learn optimal policies via trial and error. (Out of scope for this course.)

Most of this course focuses on **supervised learning**. The next few slides give each pillar a one-slide tour.

Supervised Learning



Given: training pairs (X, y) .

Goal: learn $f: X \rightarrow y$ that generalizes to new examples.

Two flavors:

y is a category → **classification** (spiral / elliptical).

y is a number → **regression** (photo-z).

Astronomy examples

Photo-z (regression) · **Galaxy morphology** (classification) · **Star–galaxy separation** (classification) · **Stellar parameters** from spectra (regression)

Unsupervised Learning

Same data, but no labels — the model has to find structure on its own.

Clustering

Group similar examples together.

Algorithms: K-Means, DBSCAN, Gaussian Mixture. Astro: spectral types, source populations.

Dimensionality reduction

Compress high-D data into a few key axes.

Algorithms: PCA, t-SNE, UMAP, autoencoders. Astro: galaxy spectra $\rightarrow$ eigenspectra.

Density estimation

Learn the probability distribution of the data.

Algorithms: KDE, GMM, normalizing flows. Astro: cosmological parameter inference.

When to reach for unsupervised: no labels yet, exploring a new dataset, or just curious what's in there.

Semi- & Self-Supervised Learning

The hybrid case: a little labeled data, plus mountains of unlabeled data.

Semi-supervised

Use unlabeled X to discover useful structure, then exploit it to fit a model on the small labeled (X, y) set.

Astro example:

*spectroscopic redshifts exist for $\sim 10^6$ galaxies;
photometry for $\sim 10^{10}$.*

Self-supervised

Manufacture labels from the data itself: *predict the next word, fill in the missing pixel, match the augmented view.*

Powers **modern foundation models** (GPT, CLIP, BERT, AstroCLIP).

Why it matters in astro: labels are scarce, raw observations are abundant. Self-supervised pre-training is becoming the standard recipe for survey data.

Reinforcement Learning (Out of Scope)

An agent interacts with an environment and learns a policy that maximizes long-term reward.



Famous successes: AlphaGo, AlphaFold's training stage, robotic control, video games.

We will not cover RL further in this course.

Problem Archetypes — A Map

The same paradigm (e.g. supervised) gives rise to several common problem types.

Classification

spiral / elliptical, star / galaxy

Regression

photo-z, stellar mass, luminosity

Clustering

find groups in unlabeled data

Image analysis

object detection, segmentation

Anomaly detection

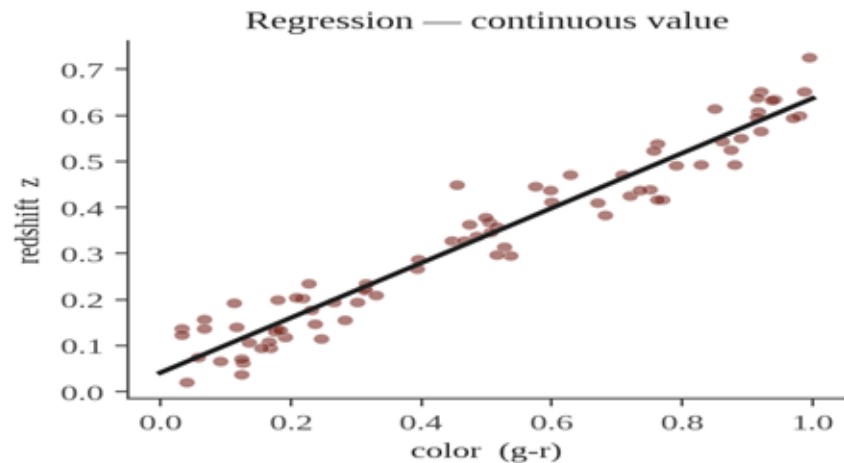
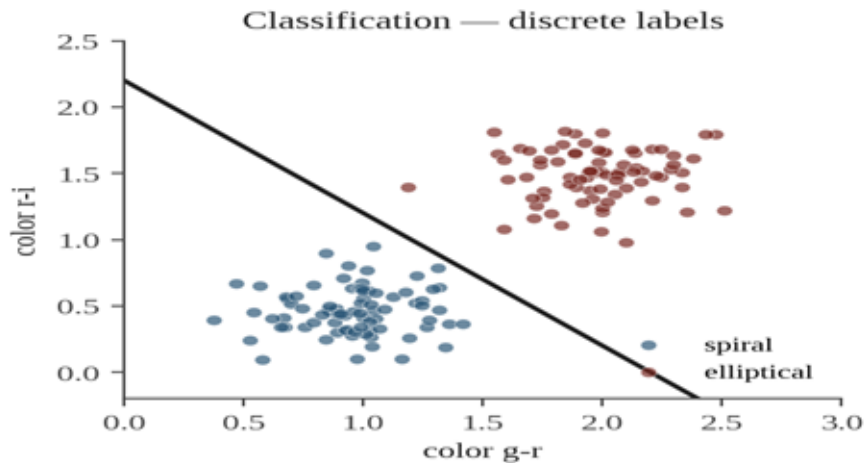
outliers from the norm

Generative

synthesize images / spectra / text

The next three slides expand the most important ones.

Classification vs Regression



Classification

Target y is a **discrete label**.

Binary: 2 classes (lens / no-lens).

Multi-class: many (galaxy morphology).

Regression

Target y is a **continuous number**.

Examples: redshift z , stellar mass, luminosity.

Metric: RMSE, bias.

Image Analysis & Generative Tasks

Beyond tabular data — the modern frontier uses pixel-level models.

Object detection

Find and box objects in an image.

*Astro: detect galaxies, transients,
lensed arcs in survey images.*

Segmentation

Label every pixel by its class.

*Astro: separate galaxy light from
cosmic-ray hits, deblending.*

Generation

Synthesize new examples from
learned distribution.

*Astro: simulate realistic galaxy
images, augment training sets,
generate spectra.*

Typical models: convolutional networks (CNNs) for detection / segmentation; *diffusion models & GANs* for generation.

Anomaly Detection & Clustering

Anomaly detection

Goal: find examples that don't match the bulk distribution.

Methods: Isolation Forest, One-Class SVM, autoencoder reconstruction error, *Z-score / IQR* (Lec 2).

Astronomy: *rare transients, gravitational lenses, weird quasars, instrument glitches.*

Clustering

Goal: group similar examples without using any labels.

Methods: K-means, DBSCAN, hierarchical, Gaussian Mixture Models.

Astronomy: *spectral type discovery, exoplanet population studies, source clustering for cosmology.*

Common thread: both work without labels — they let the data speak for itself.

Why AI/ML — Why Now?

Three trends came together in the 2010s and made modern ML inevitable.

1. Data

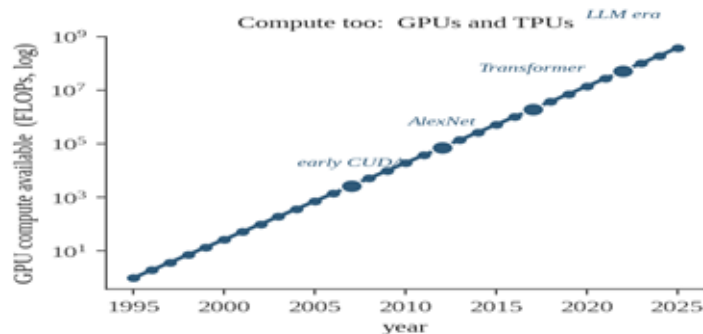
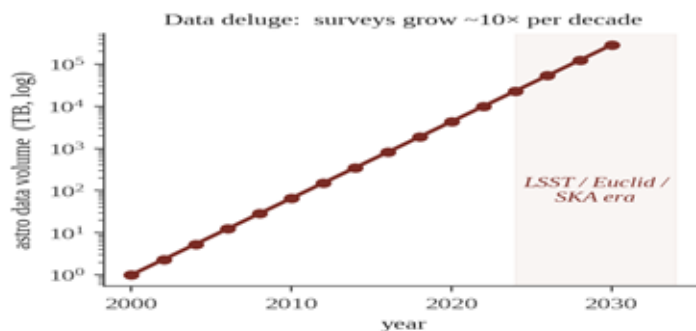
Surveys grow $\sim 10\times$ per decade —
SDSS $\rightarrow$ DES $\rightarrow$ LSST $\rightarrow$ Euclid $\rightarrow$
SKA. Multi-modal data everywhere.

2. Algorithms

Modern toolbox: backprop, CNNs,
Transformers, diffusion models,
foundation models.

3. Hardware

Massive parallel compute: NVIDIA
GPUs, Google TPUs. CUDA (2007)
democratized this.



Recap — What You Now Know

- 1 AI $\supset$ ML $\supset$ Deep Learning**
Nested concepts; the rest of this course lives in classical ML.
- 2 Four paradigms**
Supervised, Unsupervised, Semi-/Self-, Reinforcement (out of scope).
- 3 Problem archetypes**
Classification, regression, clustering, image analysis, anomaly, generative.
- 4 ML vs traditional**
Examples $\rightarrow$ rules, not the other way around.
- 5 Why now**
Data deluge, modern algorithms, parallel compute.

Next $\rightarrow$ Lecture 1b: feature vectors, data modalities, framing a problem precisely.